

STATISTICS REPORT

ACT 25MC1 - MATHEMATICS



Test Code: 25MC1

School: Sample High School

Grade: 11

Test Date: 10/31/2025

N* = 129

Number of Items = 41

Group: All Students Tested

Strand	Overall	Number and Quantity: Basic Operations & Applications (NBO)	Number and Quantity: Numbers: Concepts & Properties (NNC)	Algebra: Expressions, Equations, & Inequalities (AXE)	Functions: Functions (FFU)	Geometry: Graphical Representations (GGR)	Geometry: Properties of Plane Figures (GPP)	Geometry: Measurement (GME)	Statistics/Probability: Probability, Statistics, & Data Analysis (SPS)
Score Data									
Number of Items	41	3	4	8	6	4	3	3	10
Total Points Possible	41	3	4	8	6	4	3	3	10
Maximum Raw Score	35	3	4	8	6	4	3	3	9
Minimum Raw Score	14	0	0	0	1	0	0	0	3
Range of Raw Scores	21	3	4	8	5	4	3	3	6

Assessment Statistics

Mean Raw Score	24.8	2.0	2.5	4.6	4.0	2.0	1.7	1.8	6.2
Mean Percent Score	60.4%	65.9%	62.8%	57.5%	66.0%	49.2%	58.1%	59.2%	62.4%
Met 55% Benchmark Score (% of Students)	76.0%	73.6%	52.7%	53.5%	65.9%	35.7%	65.1%	65.9%	69.0%
Standard Deviation	3.96	0.81	0.94	1.29	1.45	1.02	0.81	0.92	1.41

Item Statistics

ITEM	STRAND CODE	BENCHMARK	STUDENT GROUP	A/F	B/G	C/H	D/J	E/K	BLANK	KEY	Difficulty Index	Discrimination Index
1	SPS Statistics/Probability: Probability, Statistics, & Data Analysis		Top 27% Students	0%	11%	3%	86%	0%	0%	D	0.60	0.46
	411	Calculate the missing data value given the average and all data values but one	All Students	6%	23%	10%	60%	0%	1%			
			Bottom 27% Students	11%	37%	9%	40%	0%	3%			
2	SPS Statistics/Probability: Probability, Statistics, & Data Analysis		Top 27% Students	3%	20%	9%	69%	0%	0%	J	0.60	0.11
	551	Use Venn diagrams in counting	All Students	7%	23%	9%	60%	0%	1%			
			Bottom 27% Students	9%	31%	3%	57%	0%	0%			
3	SPS Statistics/Probability: Probability, Statistics, & Data Analysis		Top 27% Students	3%	71%	17%	9%	0%	0%	B	0.71	-0.03
	431	Determine the probability of a simple event	All Students	6%	71%	14%	9%	0%	0%			
			Bottom 27% Students	6%	74%	11%	9%	0%	0%			

Items 7, 16, 29 and 40 are field-tested and not included in the scoring.

***Use the data with caution when N < 10.**

ITEM	STRAND CODE	BENCHMARK	STUDENT GROUP	A/F	B/G	C/H	D/J	E/K	BLANK	KEY	Difficulty Index	Discrimination Index
4	GPP Geometry: Properties of Plane Figures		Top 27% Students	49%	29%	17%	6%	0%	0%	F	0.57	-0.11
	421	Exhibit knowledge of basic angle properties and special sums of angle measures	All Students	57%	22%	14%	8%	0%	0%			
			Bottom 27% Students	60%	26%	6%	9%	0%	0%			
5	AXE Algebra: Expressions, Equations, & Inequalities		Top 27% Students	9%	14%	57%	17%	0%	3%	C	0.36	0.34
	451	Multiply two binomials	All Students	18%	20%	36%	22%	0%	4%			
			Bottom 27% Students	20%	29%	23%	26%	0%	3%			
6	NNC Number and Quantity: Numbers: Concepts & Properties		Top 27% Students	6%	17%	9%	69%	0%	0%	J	0.62	0.09
	681	Use addition, subtraction, and scalar multiplication of vectors and of matrices	All Students	6%	24%	7%	62%	0%	1%			
			Bottom 27% Students	6%	26%	9%	60%	0%	0%			
8	NNC Number and Quantity: Numbers: Concepts & Properties		Top 27% Students	3%	17%	69%	11%	0%	0%	H	0.65	0.11
	551	Work with problems involving positive integer exponents, including squares/square roots and cubes/cube roots	All Students	7%	17%	65%	10%	0%	1%			
			Bottom 27% Students	6%	23%	57%	11%	0%	3%			
9	AXE Algebra: Expressions, Equations, & Inequalities		Top 27% Students	0%	11%	6%	83%	0%	0%	D	0.60	0.43
	421	Solve routine first-degree equations	All Students	6%	23%	10%	60%	0%	1%			
			Bottom 27% Students	11%	37%	9%	40%	0%	3%			
10	NBO Number and Quantity: Basic Operations & Applications		Top 27% Students	9%	11%	63%	17%	0%	0%	H	0.62	0.09
	421	Solve two or three-step conversion problems within or between measurement systems	All Students	8%	18%	62%	12%	0%	1%			
			Bottom 27% Students	3%	29%	54%	11%	0%	3%			
11	FFU Functions: Functions		Top 27% Students	9%	74%	11%	6%	0%	0%	B	0.71	0.09
	411	Evaluate linear and quadratic functions, expressed in function notation, in integers	All Students	6%	71%	16%	8%	0%	0%			
			Bottom 27% Students	11%	66%	17%	6%	0%	0%			
12	SPS Statistics/Probability: Probability, Statistics, & Data Analysis		Top 27% Students	3%	23%	6%	69%	0%	0%	J	0.60	0.17
	531	Compute straightforward probabilities for common situations	All Students	7%	24%	9%	60%	0%	0%			
			Bottom 27% Students	14%	29%	6%	51%	0%	0%			
13	FFU Functions: Functions		Top 27% Students	71%	9%	14%	6%	0%	0%	A	0.57	0.31
	551	Express the sine/cosine/tangent of an angle in a right triangle as a ratio of side lengths	All Students	57%	22%	13%	8%	0%	0%			
			Bottom 27% Students	40%	34%	14%	11%	0%	0%			

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ITEM	STRAND CODE	BENCHMARK	STUDENT GROUP	A/F	B/G	C/H	D/J	E/K	BLANK	KEY	Difficulty Index	Discrimination Index
14	AXE Algebra: Expressions, Equations, & Inequalities		Top 27% Students	3%	17%	6%	74%	0%	0%	J	0.60	0.20
	621	Solve systems of two linear equations	All Students	5%	26%	9%	60%	0%	1%			
			Bottom 27% Students	6%	26%	11%	54%	0%	3%			
15	AXE Algebra: Expressions, Equations, & Inequalities		Top 27% Students	63%	17%	9%	11%	0%	0%	A	0.57	0.17
	531	Add, subtract, multiply and divide polynomials	All Students	57%	22%	13%	9%	0%	0%			
			Bottom 27% Students	46%	26%	11%	17%	0%	0%			
17	AXE Algebra: Expressions, Equations, & Inequalities		Top 27% Students	57%	20%	11%	11%	0%	0%	A	0.55	0.00
	431	Add and subtract simple algebraic expressions	All Students	55%	22%	15%	8%	0%	0%			
			Bottom 27% Students	57%	20%	11%	11%	0%	0%			
18	NNC Number and Quantity: Numbers: Concepts & Properties		Top 27% Students	3%	23%	17%	54%	0%	3%	J	0.63	-0.20
	571	Exhibit knowledge of the basics of complex number addition and subtraction	All Students	4%	24%	9%	63%	0%	1%			
			Bottom 27% Students	6%	17%	3%	74%	0%	0%			
19	FFU Functions: Functions		Top 27% Students	0%	94%	6%	0%	0%	0%	B	0.74	0.54
	461	Write a missing term in a given pattern that has a constant factor between terms	All Students	6%	74%	15%	5%	0%	0%			
			Bottom 27% Students	17%	40%	29%	14%	0%	0%			
20	GME Geometry: Measurement		Top 27% Students	14%	9%	69%	9%	0%	0%	H	0.62	0.17
	221	Estimate or calculate the length of a line segment based on other lengths in a geometric figure or same direction line segments	All Students	9%	18%	62%	11%	0%	1%			
			Bottom 27% Students	6%	29%	51%	11%	0%	3%			
21	GPP Geometry: Properties of Plane Figures		Top 27% Students	9%	11%	80%	0%	0%	0%	C	0.58	0.34
	541	Use Pythagorean triples to find the third length of a right triangle	All Students	26%	12%	58%	4%	0%	0%			
			Bottom 27% Students	34%	11%	46%	9%	0%	0%			
22	NBO Number and Quantity: Basic Operations & Applications		Top 27% Students	0%	94%	6%	0%	0%	0%	G	0.74	0.54
	421	Solve two or three-step conversion problems within or between measurement systems	All Students	6%	74%	15%	5%	0%	0%			
			Bottom 27% Students	17%	40%	29%	14%	0%	0%			
23	FFU Functions: Functions		Top 27% Students	3%	3%	89%	6%	0%	0%	C	0.61	0.51
	511	Evaluate polynomial functions, expressed in function notation, in integers	All Students	9%	19%	61%	11%	0%	1%			
			Bottom 27% Students	9%	34%	37%	17%	0%	3%			

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ITEM	STRAND CODE	BENCHMARK	STUDENT GROUP	A/F	B/G	C/H	D/J	E/K	BLANK	KEY	Difficulty Index	Discrimination Index
24	GGR Geometry: Graphical Representations		Top 27% Students	26%	26%	20%	26%	0%	3%	G	0.27	0.06
	681	Recognize special characteristics of parabolas and circles	All Students	21%	27%	26%	20%	0%	6%			
			Bottom 27% Students	14%	20%	31%	26%	0%	9%			
25	GGR Geometry: Graphical Representations		Top 27% Students	74%	14%	9%	3%	0%	0%	A	0.57	0.34
	673	Given an equation or function, find an equation or function whose graph is a translation by a specified amount up or down	All Students	57%	20%	14%	9%	0%	0%			
			Bottom 27% Students	40%	34%	20%	6%	0%	0%			
26	AXE Algebra: Expressions, Equations, & Inequalities		Top 27% Students	6%	63%	26%	6%	0%	0%	G	0.71	-0.11
	631	Manipulate expressions and equations	All Students	6%	71%	16%	7%	0%	0%			
			Bottom 27% Students	6%	74%	9%	11%	0%	0%			
27	SPS Statistics/Probability: Probability, Statistics, & Data Analysis		Top 27% Students	6%	74%	17%	3%	0%	0%	B	0.74	0.03
	531	Compute straightforward probabilities for common situations	All Students	6%	74%	14%	6%	0%	0%			
			Bottom 27% Students	3%	71%	17%	9%	0%	0%			
28	GME Geometry: Measurement		Top 27% Students	60%	17%	11%	11%	0%	0%	F	0.57	0.03
	521	Compute the perimeter of simple composite geometric figures with unknown side lengths	All Students	57%	21%	14%	8%	0%	0%			
			Bottom 27% Students	57%	20%	14%	9%	0%	0%			
30	GGR Geometry: Graphical Representations		Top 27% Students	17%	0%	0%	83%	0%	0%	J	0.55	0.57
	431	Locate points in the coordinate plane	All Students	21%	12%	12%	55%	0%	1%			
			Bottom 27% Students	26%	26%	23%	26%	0%	0%			
31	GPP Geometry: Properties of Plane Figures		Top 27% Students	0%	6%	89%	6%	0%	0%	C	0.60	0.57
	651	Apply properties of 30°-60°-90°, 45°-45°-90°, similar, and congruent triangles	All Students	5%	22%	60%	12%	0%	1%			
			Bottom 27% Students	14%	40%	31%	11%	0%	3%			
32	NBO Number and Quantity: Basic Operations & Applications		Top 27% Students	6%	3%	0%	91%	0%	0%	J	0.62	0.37
	411	Solve routine two or three-step arithmetic problems	All Students	5%	23%	9%	62%	0%	1%			
			Bottom 27% Students	3%	31%	11%	54%	0%	0%			
33	AXE Algebra: Expressions, Equations, & Inequalities		Top 27% Students	3%	20%	69%	9%	0%	0%	C	0.65	0.14
	521	Solve real-world problems by using first-degree equations	All Students	8%	16%	65%	10%	0%	1%			
			Bottom 27% Students	14%	23%	54%	9%	0%	0%			

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34	FFU Functions: Functions		Top 27% Students	0%	94%	6%	0%	0%	0%	G	0.74	0.54
	561	Find the next term in a sequence described recursively	All Students	6%	74%	15%	5%	0%	0%			
			Bottom 27% Students	17%	40%	29%	14%	0%	0%			
35	GGR Geometry: Graphical Representations		Top 27% Students	23%	6%	66%	6%	0%	0%	C	0.57	0.26
	311	Locate both positive and negative rational numbers on the number line	All Students	26%	12%	57%	5%	0%	0%			
			Bottom 27% Students	37%	17%	40%	6%	0%	0%			
36	SPS Statistics/Probability: Probability, Statistics, & Data Analysis		Top 27% Students	6%	3%	0%	91%	0%	0%	J	0.61	0.40
	531	Compute straightforward probabilities for common situations	All Students	6%	24%	9%	61%	0%	0%			
			Bottom 27% Students	3%	34%	11%	51%	0%	0%			
37	AXE Algebra: Expressions, Equations, & Inequalities		Top 27% Students	26%	3%	60%	11%	0%	0%	C	0.57	0.17
	652	Factor polynomial equations to find roots or zeros	All Students	29%	9%	57%	5%	0%	0%			
			Bottom 27% Students	34%	23%	43%	0%	0%	0%			
38	SPS Statistics/Probability: Probability, Statistics, & Data Analysis		Top 27% Students	3%	23%	6%	69%	0%	0%	J	0.60	0.11
	511	Calculate the average given the frequency counts of all the data values	All Students	5%	25%	9%	60%	0%	1%			
			Bottom 27% Students	0%	31%	11%	57%	0%	0%			
39	SPS Statistics/Probability: Probability, Statistics, & Data Analysis		Top 27% Students	9%	11%	74%	6%	0%	0%	C	0.64	0.17
	511	Calculate the average given the frequency counts of all the data values	All Students	8%	18%	64%	10%	0%	1%			
			Bottom 27% Students	14%	23%	57%	3%	0%	3%			
41	NNC Number and Quantity: Numbers: Concepts & Properties		Top 27% Students	3%	23%	6%	69%	0%	0%	D	0.61	0.17
	511	Order fractions	All Students	5%	23%	9%	61%	0%	2%			
			Bottom 27% Students	14%	26%	6%	51%	0%	3%			
42	SPS Statistics/Probability: Probability, Statistics, & Data Analysis		Top 27% Students	71%	9%	14%	6%	0%	0%	F	0.57	0.31
	612	Exhibit knowledge of, calculate, or use the standard deviation	All Students	57%	22%	14%	7%	0%	0%			
			Bottom 27% Students	40%	34%	17%	9%	0%	0%			
43	GME Geometry: Measurement		Top 27% Students	23%	3%	71%	3%	0%	0%	C	0.58	0.31
	611	Use relationships involving area, perimeter, and volume to compute another measure	All Students	27%	9%	58%	5%	0%	0%			
			Bottom 27% Students	37%	17%	40%	6%	0%	0%			

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44	FFU Functions: Functions		Top 27% Students	0%	14%	3%	83%	0%	0%	J	0.60	0.40
	611	Evaluate composite functions at integer values	All Students	6%	24%	9%	60%	0%	0%			
			Bottom 27% Students	11%	37%	9%	43%	0%	0%			
45	SPS Statistics/Probability: Probability, Statistics, & Data Analysis		Top 27% Students	71%	9%	14%	6%	0%	0%	A	0.57	0.31
	631	Compute a probability when the event and/or sample space are not given or obvious	All Students	57%	22%	14%	7%	0%	0%			
			Bottom 27% Students	40%	34%	17%	9%	0%	0%			

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